rag -- 2026-11-06

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P3] Reasoning over Non-Parametric Memory in Multilingual LLMs: A Cognitive-Based Analysis of Relational Knowledge -- REL-BLOOM provides the most systematic evaluation of how retrieved context is actually used for relational reasoning, exposing a key blind spot in current RAG research that informs both grounding and system design.

Themes

Several papers target the core problem of improving the grounding of generated answers by enriching the retrieved context. CacheGraft fuses KV-caches from a large "sharer" model into a small "receiver" model's latent space, achieving up to 8 percentage-point gains on QA benchmarks but at the cost of massive offline storage for per-document caches [P1]. In a different domain, HERIT augments RAG with glossaries, Korean translations, and a two-stage Proposer-Fusor ensemble to produce high-quality pseudo-labels for historical Hanja-to-English translation, pushing BLEU past 38 points while still relying on a single LLM for both stages [P2].

A complementary line of work examines how well current RAG systems actually reason over retrieved evidence. REL-BLOOM introduces a multilingual diagnostic suite that isolates relational reasoning across Bloom's cognitive levels, revealing steep drops--from Apply to Analyze--across 43 models and highlighting that high end-to-end scores on standard QA do not guarantee deeper understanding [P3]. Meanwhile, Signed but Steerable investigates the integrity of offline retrieval packs, showing that cosine-tolerance checks can leave up to 5.5 % of queries vulnerable and that attacks can halve correct answers on edge devices, underscoring a security-related evaluation gap for RAG deployments [P4].

System-design innovations also appear. ZeroRAG replaces centralized indexes with a semantic peer-routing overlay that self-organises using lightweight neighbor advertisements and scales to 100 k peers [P5]. Across the set, the papers converge on the need to balance retrieval quality, grounding fidelity, and practical constraints (storage, security, decentralisation), while diverging in focus--some on latent-space enrichment, others on benchmark diagnostics or network architecture.

5 theme sentence(s) checked against the paper summaries; 1 corrected, 0 removed.

At a glance

[P1] CacheGraft: Enhancing Retrieval-Augmented Generation via Latent-Space Document Enrichment · relevance 0.46
CacheGraft enriches retrieval-augmented generation by offline fusing large-model KV caches into a small receiver, yielding consistent answer quality gains but requiring substantial storage.

[P2] HERIT: Enabling Global Access to Hanja-to-English Historical Translation by Mitigating Temporal Bias · relevance 0.35
HERIT is a retrieval-augmented, two-stage translation pipeline that generates high-quality pseudo-labels for Hanja-to-English translation, achieving BLEU scores up to 38.5 and surpassing existing baselines, but its evaluation is limited to a single LLM and document set.

[P3] Reasoning over Non-Parametric Memory in Multilingual LLMs: A Cognitive-Based Analysis of Relational Knowledge · relevance 0.35
REL-BLOOM is a multilingual diagnostic benchmark of 450 multiple-choice questions that reveals large performance drops on higher-order relational reasoning despite near-perfect scores on lower-order tasks.

[P4] Signed but Steerable: Measuring and Closing the Text-Vector Integrity Gap in Offline Retrieval-Augmented Generation · relevance 0.34
Signed knowledge packs for offline RAG can be verified, but any tolerance-based check leaves a predictable exposure (up to 5.5% of queries, or 98-100% with INT8-calibrated thresholds) that attackers can exploit to halve correct answers on a Raspberry Pi device.

[P5] ZeroRAG: Adaptive Semantic Peer Routing for Decentralized Retrieval-Augmented Generation · relevance 0.33
ZeroRAG is a decentralized routing layer for retrieval-augmented generation that improves recall and RAG accuracy without central indexes, scaling to 100 000 peers.

[P1] CacheGraft: Enhancing Retrieval-Augmented Generation via Latent-Space Document Enrichment

Vihan Lakshman, Peter Baile Chen, Zhuohan Gu, Michael Cafarella · arXiv 2611.07393v1 · https://arxiv.org/abs/2611.07393v1 · relevance 0.46

TL;DR: CacheGraft enriches retrieval-augmented generation by offline fusing large-model KV caches into a small receiver, yielding consistent answer quality gains but requiring substantial storage.

Problem
CacheGraft improves answer quality over a text-only receiver baseline that uses raw textual passages [1]. There is a need for a compute-efficient way to provide richer contextual knowledge to a small model without incurring the cost of running a large model online [1][2].

Method
CacheGraft builds on the Cache-to-Cache (C2C) fusion framework: a 4 B-parameter "sharer" model and a 0.6 B "receiver" model each pre-fill their KV caches on the same passage; a lightweight per-layer fuser network merges the sharer's cache into the receiver's latent space, producing a fused KV cache that can be stored offline and injected at query time [2][3]. The receiver then generates answers alone, using only the fused cache and the query text [3].

Key results
Across five QA benchmarks, CacheGraft improves answer quality over the receiver-only baseline in 29 of 30 comparisons, with 17 of those gains statistically significant [1][4]. Gains of up to 8 percentage points are reported, e.g., AdversarialQA (top-1 recall +6.0 %*, oracle +7.2 %*), SQuAD (retrieved +7.9 %*, oracle +6.4 %*), CovidQA (retrieved +3.8 %*, oracle +5.4 %*), TriviaQA (retrieved +3.5 %*, oracle +3.2 %*), and QASPER (oracle +2.9 %*) [4]. Query-time latency increases from 334 ms (receiver-only) to 574 ms for CacheGraft, with modest GPU memory growth (1.22 GB -> 1.27 GB) [6].

Limitations
The main practical limitation is storage: storing a KV cache per document can require orders of magnitude more space than raw text (~ 1 TB for 100 K documents) [5]. The study only evaluates single-document retrieval and a single off-the-shelf fuser model, leaving top-k retrieval and multi-model extensions untested [5]. Additionally, CacheGraft incurs higher inference latency compared with plain text-RAG [6].

Sources

[1] arXiv 2611.07393v1.pdf, arXiv abstract

[2] arXiv 2611.07393v1.pdf, page 1

[3] arXiv 2611.07393v1.pdf, page 2

[4] arXiv 2611.07393v1.pdf, page 3

[5] arXiv 2611.07393v1.pdf, page 3

[6] arXiv 2611.07393v1.pdf, page 4

[P2] HERIT: Enabling Global Access to Hanja-to-English Historical Translation by Mitigating Temporal Bias

Dayeon Jeong, Kyeongpil Kang · arXiv 2611.04949v1 · https://arxiv.org/abs/2611.04949v1 · relevance 0.35

TL;DR: HERIT is a retrieval-augmented, two-stage translation pipeline that generates high-quality pseudo-labels for Hanja-to-English translation, achieving BLEU scores up to 38.5 and surpassing existing baselines, but its evaluation is limited to a single LLM and document set.

Problem
Historical Hanja archives are valuable yet inaccessible because manual translation is costly and existing machine translation models suffer from temporal bias and a lack of parallel Hanja-English data [1]. Prior attempts using scarce Hanja-English corpora or multilingual models remain limited by data scarcity [2].

Method
HERIT builds a domain-adaptation pipeline that (i) uses retrieval-augmented generation (RAG) to create pseudo-labels from abundant Hanja-Korean corpora, (ii) enriches prompts with glossaries and Korean translations, (iii) applies few-shot examples (static and dynamic) during inference, and (iv) employs a two-stage Proposer-Fusor architecture that ensembles multiple draft translations into a final output [1][2][3][4][5].

Key results
- Using Gemini-2.5-Flash as a baseline, adding a glossary raises BLEU from 15.720 to 23.747, METEOR from 0.421 to 0.504, and chrF++ from 43.971 to 51.478 [5].
- Incorporating Korean inputs further improves BLEU to 25.457, METEOR to 0.520, and chrF++ to 52.637 [5].
- Adding RAG with dynamic few-shot shots (e.g., Hanja + Korean) pushes BLEU to 36.430, METEOR to 0.606, and chrF++ to 60.545 [5].
- The full Proposer-Fusor configuration (Proposer Kor, Fusor Hanja + Eng) attains the highest BLEU of 38.462, METEOR of 0.625, and chrF++ of 62.501 [5].
- These scores exceed those of single-stage baselines and other LLMs listed in Table 2 (e.g., Sonnet-4.5 achieves BLEU ~ 18.1) [3].
- Human and LLM-based evaluations confirm HERIT's superiority over strong baselines and its ability to mitigate temporal overfitting [6].

Limitations
The study used only Gemini-2.5-Flash for both Proposer and Fusor and focused exclusively on AJD documents for pseudo-labeling due to budget constraints [7]. Maximum input length was limited to 2,048 tokens (~ 800 Hanja characters), causing ~ 6.7 % of training data to be filtered out and requiring chunking for longer documents [7]. Expanding to diverse LLMs, additional document types (e.g., JRS), and longer sequences may further improve performance [7].

Sources

[1] arXiv 2611.04949v1.pdf, arXiv abstract

[2] arXiv 2611.04949v1.pdf, page 2

[3] arXiv 2611.04949v1.pdf, page 4

[4] arXiv 2611.04949v1.pdf, page 5

[5] arXiv 2611.04949v1.pdf, page 6

[6] arXiv 2611.04949v1.pdf, page 9

[7] arXiv 2611.04949v1.pdf, page 10

[P3] Reasoning over Non-Parametric Memory in Multilingual LLMs: A Cognitive-Based Analysis of Relational Knowledge

Zohreh Abbasi Liasi, Batoul Hammoud, Anas Madkoor, Ehsaneddin Asgari · arXiv 2611.06715v1 · https://arxiv.org/abs/2611.06715v1 · relevance 0.35

TL;DR: REL-BLOOM is a multilingual diagnostic benchmark of 450 multiple-choice questions that reveals large performance drops on higher-order relational reasoning despite near-perfect scores on lower-order tasks.

Problem - Retrieval-augmented generation (RAG) assumes language models can reason over relational evidence supplied in the prompt, yet existing benchmarks only test whether retrieved context is used, not whether the specific relation is understood [1][2].

Method - The authors built REL-BLOOM from 30 curated Wikidata relations, instantiating each at five Bloom-inspired cognitive levels (Remember, Understand, Apply, Analyze, Evaluate) and three veracity conditions (Factual, Counterfactual, Neutralized), yielding 450 English multiple-choice items (30 × 5 × 3) [1][2]. These items were translated into six languages (Arabic, Chinese, English, Hindi, Persian, Spanish) and evaluated in two multilingual settings (fully translated and hybrid cross-lingual), producing 226,800 predictions from 43 models (41 open-weight across seven families plus two closed-weight Gemini models) [1][2].

Key results - Across all open-weight series, accuracy drops sharply from Apply to Analyze; the Large Granite bucket shows a 47.2-point drop [3][1]. Gemini 3.1 Pro attains 99.5 % accuracy on the four lower Bloom levels but only 60.7 % on Evaluate [1]. Scaling gains are evident: Qwen2.5 Instruct improves by 23.8 percentage points from Small to Medium and by 6.1 points from Medium to Large [3]. Apply consistently exceeds Analyze in every open-weight series [3]. Counterfactual contexts are generally harder than Factual ones, while neutralized (entity-removed) contexts can boost higher-level performance in some regimes [1].

Limitations - REL-BLOOM prioritises controlled comparison over breadth, using only one source triple per relation (30 relation clusters), which limits power for detecting small effects and generalising to unseen triples [4]. The benchmark does not measure end-to-end RAG with noisy retrieval or open-ended generation, and item-construction choices may influence level-wise gaps; memorisation, size-comparison descriptiveness, and shared templates could inflate cross-language correlations [4].

Sources

[1] arXiv 2611.06715v1.pdf, arXiv abstract

[2] arXiv 2611.06715v1.pdf, page 4

[3] arXiv 2611.06715v1.pdf, page 7

[4] arXiv 2611.06715v1.pdf, page 10

[P4] Signed but Steerable: Measuring and Closing the Text-Vector Integrity Gap in Offline Retrieval-Augmented Generation

Georgios Feretzakis, Alexandros Papaspyridis · arXiv 2611.06498v1 · https://arxiv.org/abs/2611.06498v1 · relevance 0.34

TL;DR: Signed knowledge packs for offline RAG can be verified, but any tolerance-based check leaves a predictable exposure (up to 5.5% of queries, or 98-100% with INT8-calibrated thresholds) that attackers can exploit to halve correct answers on a Raspberry Pi device.

Problem
Offline retrieval-augmented generation on edge devices relies on signed knowledge packs that contain

both text and precomputed embedding vectors, yet a signature only proves provenance, not that the vectors faithfully encode the text, creating a "text-vector integrity gap" [1].

Method
The authors evaluate a verifier that recomputes vectors and accepts them within a cosine tolerance, deriving a "margin law" that predicts hijack rates from angular retrieval margins [1]. Experiments use a Raspberry Pi 5 (CPU-only) running a local LLM, a desktop for large-scale tests, and an Apple M2 for cross-device verification; embedding models include BGE-M3 and MiniLM-L6-v2, and retrieval is performed with FAISS product-quantisation indexes [3]. They also implement code-level verification of quantisation codes with canonicalisation and a calibrated guard band [4].

Key results
- In five corpora and two embedding models, margins are small enough that an idealised floating-point-noise tolerance leaves up to 5.5 % of queries exposed (at most 1.2 % once exact duplicate documents are excluded) [1].
- A tolerance calibrated from a CPU INT8-FP32 comparison (the 0.1 % quantile of honest cosines) leaves 98-100 % of queries exposed [1].
- On a Raspberry Pi 5, an attack that respects the INT8-level tolerance reduces correct answers from 0.75 to 0.39 and raises refusals from 0.13 to 0.50 [5].
- In a FAISS product-quantisation index over 57,638 documents, tampered vectors that pass an angular-tolerance audit cause the attacker's document to rank first for 25 % of targets (95 % CI 16-37 %) [4].
- Code-level verification with canonicalisation detects and removes all 60 tampered codes in the authors' samples across three platforms [4].
- Encoder load for re-indexing reduces decoding throughput by -64.5 % to -73.9 % and periodic verification adds -22.9 % to -31.4 % throughput loss, with p99 inter-token latency increasing +108 % to +167 % [5].
- Pack-update testing (19 cases) matched expected outcomes in 17 cases; all 12 structural attacks were rejected, while two content-level screens behaved unexpectedly [5].

Limitations
The study's internal validity rests on a single reference verifier (subject to a bug fix and library change) and reported latencies exclude prompt-cache hits [6]. Construct validity is limited to the authors' definition of hijacking (gold-unit displacement) and a greedy FAISS attack that is a lower bound [6]. External validity is constrained to two English embedding models, corpora up to 57,638 documents, flat FAISS indexes, one edge board, and a single local LLM; other quantisation schemes, accelerators, or larger datasets may exhibit different tolerances and exposures [6].

Sources

[1] arXiv 2611.06498v1.pdf, arXiv abstract

[3] arXiv 2611.06498v1.pdf, page 6

[4] arXiv 2611.06498v1.pdf, page 8

[5] arXiv 2611.06498v1.pdf, page 11

[6] arXiv 2611.06498v1.pdf, page 15

[P5] ZeroRAG: Adaptive Semantic Peer Routing for Decentralized Retrieval-Augmented Generation

Petru Neague, Quinten Stokkink, Martijn de Vos, Rafael Pires et al. · arXiv 2611.07928v1 · https://arxiv.org/abs/2611.07928v1 · relevance 0.33

TL;DR: ZeroRAG is a decentralized routing layer for retrieval-augmented generation that improves recall and RAG accuracy without central indexes, scaling to 100 000 peers.

Problem
Current RAG systems rely on centralized document stores, which conflict with privacy, copyright, and regulatory constraints because owners cannot surrender custody of their data [1]. Existing decentralized retrieval approaches either re-introduce a coordinating authority or depend on global data structures such as DHTs that require continuous network-wide coordination [1][2].

Method
ZeroRAG introduces a semantic peer-routing overlay where each peer keeps only lightweight local state (neighbor advertisements, a query cache, and a bounded neighbor list) and routes queries via a best-first walk that progressively rewires the overlay toward semantically relevant neighbors [1][3]. Queries are embedded with a sentence-transformer and routed using cosine similarity scores; the network self-organizes as peers cache recent queries and adjust neighbor connections [3].

Key results
- Evaluation uses three real-world workloads: AOL-IA (~2.17 M queries from the top 10 k users), Amazon Electronics (63 001 product descriptions), and PubMedQA (biomedical Q&A) [5].
- On the Amazon Electronics workload ZeroRAG retrieves 94.8 %-98.9 % of available relevant products across all relevance bins, outperforming DHT-LSH (90.3 %-91.7 % with w=3.0, k=10, L=50) and Semantica (up to 66.7 %) [6].
- In end-to-end RAG on PubMedQA, a centralized HNSW index achieves 97.3 % recall and 79 % answer accuracy, while ZeroRAG reaches 93.0 % recall and 77 % accuracy with k=5 retrieved documents [6].
- Recall@1 improves with larger neighbor sets: from 98.614 % (?n=3) to 99.675 % (?n=100), while the average number of requests drops from 6.20 to 3.30 (~46.5 % reduction) [7].
- Parallel query fan-out also affects performance: fan-out 3 (10 rounds) yields 99.374 % recall with 1.72 rounds on average; fan-out 6 (5 rounds) gives 99.203 % recall with 1.51 rounds; fan-out 10 (3 rounds) slightly lowers recall to 97.456 % [7].
- The system self-heals after a catastrophic failure where 50 % of peers leave, maintaining recall (as shown in Figure 6) [6].

Limitations
The excerpts do not provide quantitative measurements of latency, bandwidth overhead, or detailed privacy guarantees, nor do they report results beyond 100 000 peers or on heterogeneous network conditions. Consequently, a reader cannot assess performance under those aspects from the available information.

Sources

[1] arXiv 2611.07928v1.pdf, arXiv abstract

[2] arXiv 2611.07928v1.pdf, page 3

[3] arXiv 2611.07928v1.pdf, page 5

[5] arXiv 2611.07928v1.pdf, page 7

[6] arXiv 2611.07928v1.pdf, page 9

[7] arXiv 2611.07928v1.pdf, page 11